

# abtestkit

```
library(abtestkit)
```

## What this package does

LINK TO GITHUB: <https://github.com/mbsperry-school/abtestkit>

abtestkit is a toolkit for planning and analyzing binary-outcome A/B tests. An A/B test is an experiment that is used to compare the results of a randomized experiment where subjects can get one of two experiences. It is frequently used in software, where users are given two different experiences on a website and employees are trying to answer the question: “Which is better?” Any time a big company releases a software update, it can be safely assumed that they ran some variation of an A/B test before to get the confidence needed to ship to their whole user base. Scientists and engineers are able to have confidence in the changes in the software by using sound statistical methods that measure an effect on a small group of users, that would generalize to the entire base of users. It covers both frequentist and Bayesian approaches side by side, aimed at someone who understands what an A/B test is but has never worked with Bayesian methods.

Most existing tooling makes you decide between frequentist and bayesian approaches. Comparing the two means putting together different packages that have different argument names and processes, which is confusing and discourages comparison. However, both of these methods are useful for different audiences, and the working statistician needs their coworkers and boss to readily understand their work in addition to being able to explain to non-technical audiences. Everyone that takes an intro-level statistics class is familiar with a p-value, which is the foundation of frequentist statistics. But the translation of bayesian statistical methods to stakeholders is much more straightforward. With this package, you don’t have to choose, and through inspecting the functions, you can better understand for yourself what the differences are between these schools of thought. There are two phases: planning (before the experiment) and analysis (after the experiment). This vignette walks through both using a real marketing dataset.

## The dataset

The data comes from a Kaggle marketing A/B test where 588,101 users were exposed to either an advertisement or a public service announcement (PSA) placeholder. The outcome is whether the user converted (made a purchase).

We split this into two pieces:

- **historical\_data**: 1,000 PSA users treated as pre-experiment baseline data. This is what we “know” before the experiment starts.
- **experiment\_data**: 10,000 users per arm sampled from the remaining data. This is the actual experiment.

```
data(historical_data)
data(experiment_data)
```

## Phase 1: Pre-experiment planning

Before running an experiment, you need to answer two questions: - how many users do I need - what do I believe going in?

The frequentist side answers the first question with a sample size calculation. The Bayesian side answers the second by turning your historical data into a prior distribution. `pre_experiment_comparison` handles both at once. You pass in your historical data, the column with the binary outcome, and the minimum detectable effect (the smallest change worth detecting). Here we set `mde = 0.01`, meaning we want to detect a 1 percentage point lift.

```
pre_experiment_data <- pre_experiment_comparison(  
  mde = 0.01,  
  data = historical_data,  
  outcome_column_name = "converted"  
)
```

The function returns two things: the sample size needed per group (frequentist) and a Beta prior (Bayesian).

```
n <- pre_experiment_data$n  
experiment_prior <- pre_experiment_data$experiment_prior  
  
cat("Required sample size per group:", n, "\n")  
#> Required sample size per group: 4569  
cat("Prior alpha:", round(experiment_prior$alpha, 3), "\n")  
#> Prior alpha: 0.354  
cat("Prior beta:", round(experiment_prior$beta, 3), "\n")  
#> Prior beta: 10.73  
cat("Prior method:", experiment_prior$method, "\n")  
#> Prior method: quantile
```

The sample size tells you how long to run the test. The prior encodes what you knew about conversion before seeing any experiment data. Because our historical conversion rate is low (below 50%), the prior was fit using quantile matching, which handles skewed distributions better than moment matching.

## Phase 2: Post-experiment analysis

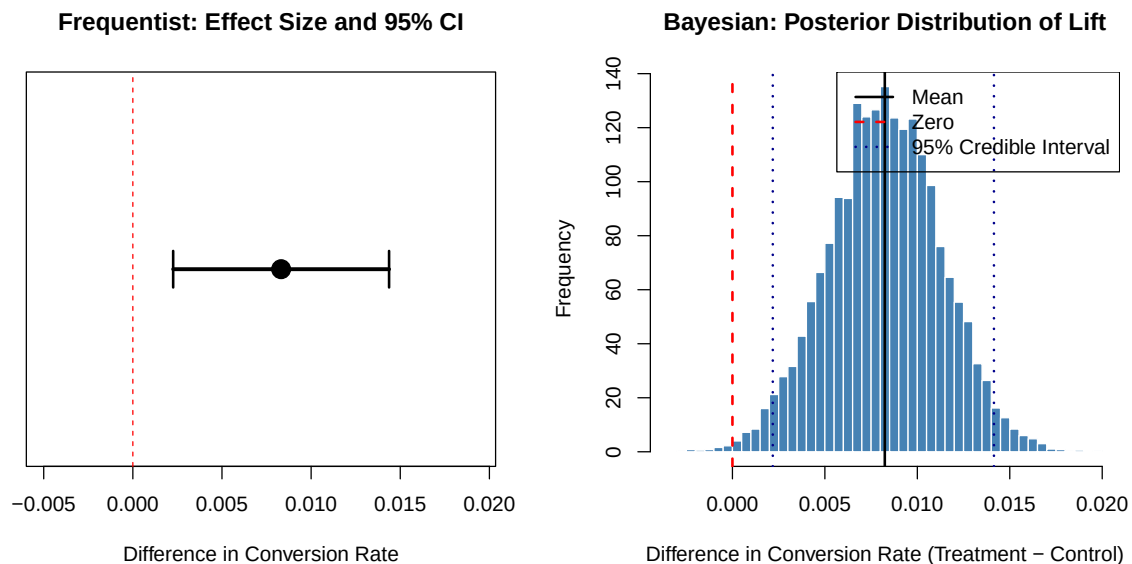
Now we have experiment results. `post_experiment_comparison` runs both analyses on the same data. The frequentist side runs a two-proportion z-test. The Bayesian side updates the prior with each group's observed conversions and compares the resulting posterior distributions.

```
post_experiment_data <- post_experiment_comparison(  
  data = experiment_data,  
  outcome_column_name = "converted",  
  group_column_name = "test.group",  
  control_value = "psa",  
  treatment_value = "ad",  
  n = n,  
  experiment_prior = experiment_prior  
)
```

## Results

`full_comparison` puts everything together with side-by-side visuals and a clear interpretation of what both methods found.

```
full_comparison(  
  freq_results = post_experiment_data$freq_results,  
  bayes_results = post_experiment_data$bayes_results,  
  posterior_control = post_experiment_data$posterior_control,  
  posterior_treatment = post_experiment_data$posterior_treatment  
)
```



```
#> === A/B Test Results Interpretation ===  
#>  
#> -- Frequentist Results --  
#> Observed effect: 0.0083  
#> p-value: 0.0071  
#> 95% Confidence Interval: [0.0023, 0.0144]  
#> The frequentist confidence interval says: if we repeated this  
#> experiment many times, 95% of the intervals we compute would  
#> contain the true effect.  
#>  
#> The result IS statistically significant at alpha = 0.05  
#> This means: if there were truly no difference between groups,  
#> we would see data this extreme less than 5% of the time.  
#> We reject the null hypothesis that the groups are the same.  
#>  
#> -- Bayesian Results --  
#> P(treatment > control): 0.9956  
#> Expected lift: 0.0083  
#> 95% Credible Interval for lift: [0.0022, 0.0143]  
#> The Bayesian credible interval says: given our prior beliefs and  
#> the observed data, there is a 95% probability that the true
```

```

#> effect falls inside this interval.
#>
#> The Bayesian analysis gives strong evidence that treatment is
#> better. There is a 99.6% probability that the treatment group
#> has a higher conversion rate.
#>
#> -- Do the two methods agree? --
#>
#> Yes. Both methods agree that the treatment group performed better.
#> The frequentist test found a significant positive effect, and the
#> Bayesian analysis assigns a high probability to treatment being
#> better.

```

The left plot shows the frequentist confidence interval for the difference in conversion rates. If the interval excludes zero (the red dashed line), the result is statistically significant. The right plot shows the Bayesian posterior distribution of the lift. The black line is the expected lift, the dashed blue lines mark the 95% credible interval, and the red dashed line is zero. If most of the distribution sits to the right of zero, the Bayesian analysis favors the treatment. The printed interpretation below the plots explains what each method concluded, whether they agree, and the difference between a confidence interval and a credible interval.

## Conclusions

You should not have to choose a school of thought to run one experiment. The frequentist result gives you the significance threshold that a review process or a skeptical colleague will expect. The Bayesian result gives you the probability that the treatment is better, which is the quantity a non-technical stakeholder was asking about in the first place. Producing both from one set of inputs means the comparison is easy.

If you are comfortable with p-values and have been meaning to find out what Bayesian methods would add, running your next test through `pre_experiment_comparison()` and `full_comparison()` is the cheapest possible way to find out. You get the analysis you would have run anyway, plus a second answer to check it against.